Table 1: Results for RNNs using the same settings as in the main paper: R^2, coset clustering score ($s_{cluster}$), and coset disjointness, averaged across 300 seeds. Note that replacing each neuron's preactivation (layerwise) by its fitted sinusoid(s) yields 100% test accuracy in all runs, and rotation traces form coherent loops with near-monotone angular progression ($|\rho| = 0.9968 \pm 0.0189$).

$s_{cluster}$	coset disjointness	R^2
0.9968 ± 0.0032	0.9999 ± 0.0009	0.9934 ± 0.0078

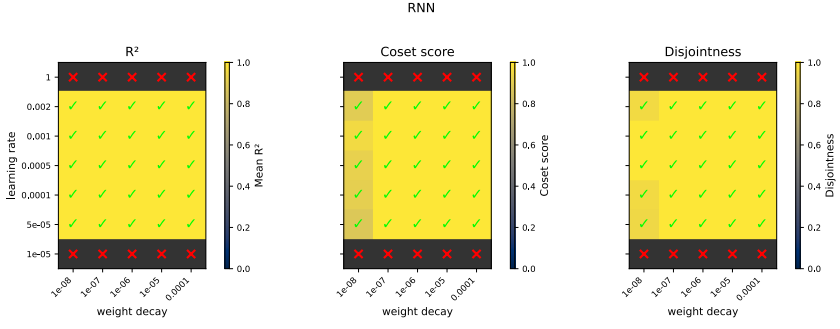

Figure 1: Hyperparameter tuning results for RNN. Each heatmap shows a sweep over learning rate (y-axis) and weight decay (x-axis). Columns correspond to different metrics: the first column reports the mean coefficient of determination R^2 between network preactivations and the sinusoidal fits, the second reports the coset clustering score, and the third reports the disjointness metric. Colors indicate the metric value averaged over random seeds, we only fill a cell when all seeds reach 100% train and test accuracy during training; runs where any seed fails to generalize to 100% accuracy are marked with a red "×". Among successful runs, a purple "○" denotes that replacing the neuron preactivations by the sinusoidal fits results in test accuracy $< 99.9999\%$, while a green "✓" denotes replace neuron's activation by the sinusoidal fit results in test accuracy $> 99.9999\%$.